

Hellenic Complex Systems Laboratory

Uncertainty of Measurement and Diagnostic Accuracy Measures

Technical Report VII

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Uncertainty of Measurement and Diagnostic Accuracy Measures

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Short Description of the Demonstration

This Demonstration compares various diagnostic accuracy measures of two diagnostic tests. The two tests measure the same measurand, in normally distributed nondiseased and diseased populations, for various values of the prevalence of the disease, of the population means and standard deviations, and of the tests' measurement uncertainty. A normal distribution of the uncertainty is assumed. The mean and the standard deviation of each population and the uncertainty of each test are measured in arbitrary units. The measures compared are the positive predictive value (*PPV*), the negative predictive value (*NPV*), the (diagnostic) odds ratio (*OR*), the likelihood ratio for a positive test result (*LR+*), and the likelihood ratio for a negative test result (*LR-*). The measures are calculated versus the sensitivity or the specificity of each test. This can be selected by clicking the respective button. The available plot types are: both measures (first test: blue plot, second test: orange plot), partial derivatives of both measures with respect to measurement uncertainty (first test: blue plot, second test: orange plot), difference, and ratio of the two measures.

Search Terms

uncertainty of measurement, sensitivity, specificity, diagnostic test, clinical accuracy, diagnostic accuracy, permissible uncertainty, normal distribution, binormal distribution, positive predictive value, negative predictive value, likelihood ratio, odds ratio

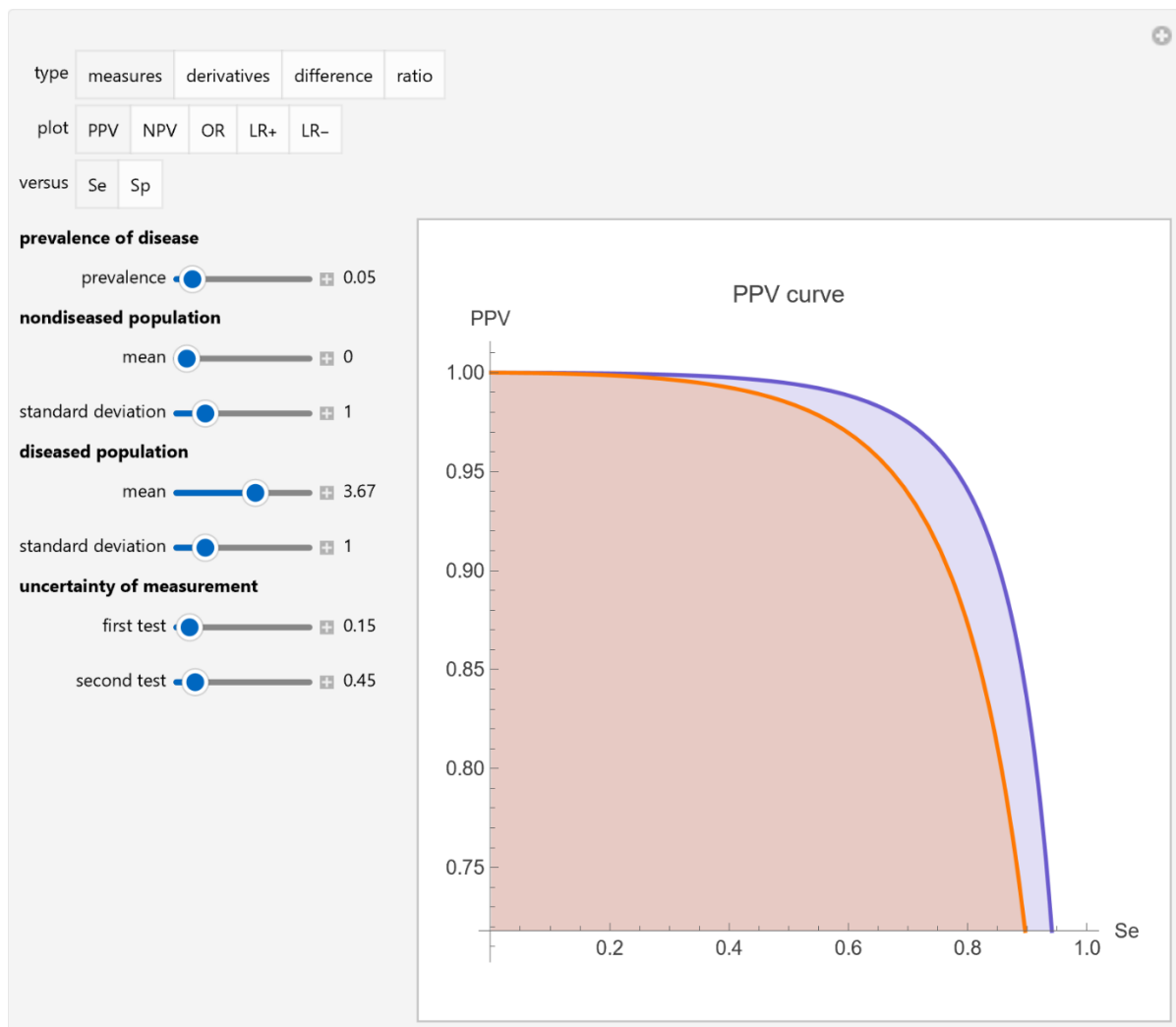


Figure 1: Positive predictive value vs sensitivity curve plots of two diagnostic tests (first test: blue plot, second test: orange plot), with the settings shown at the left.

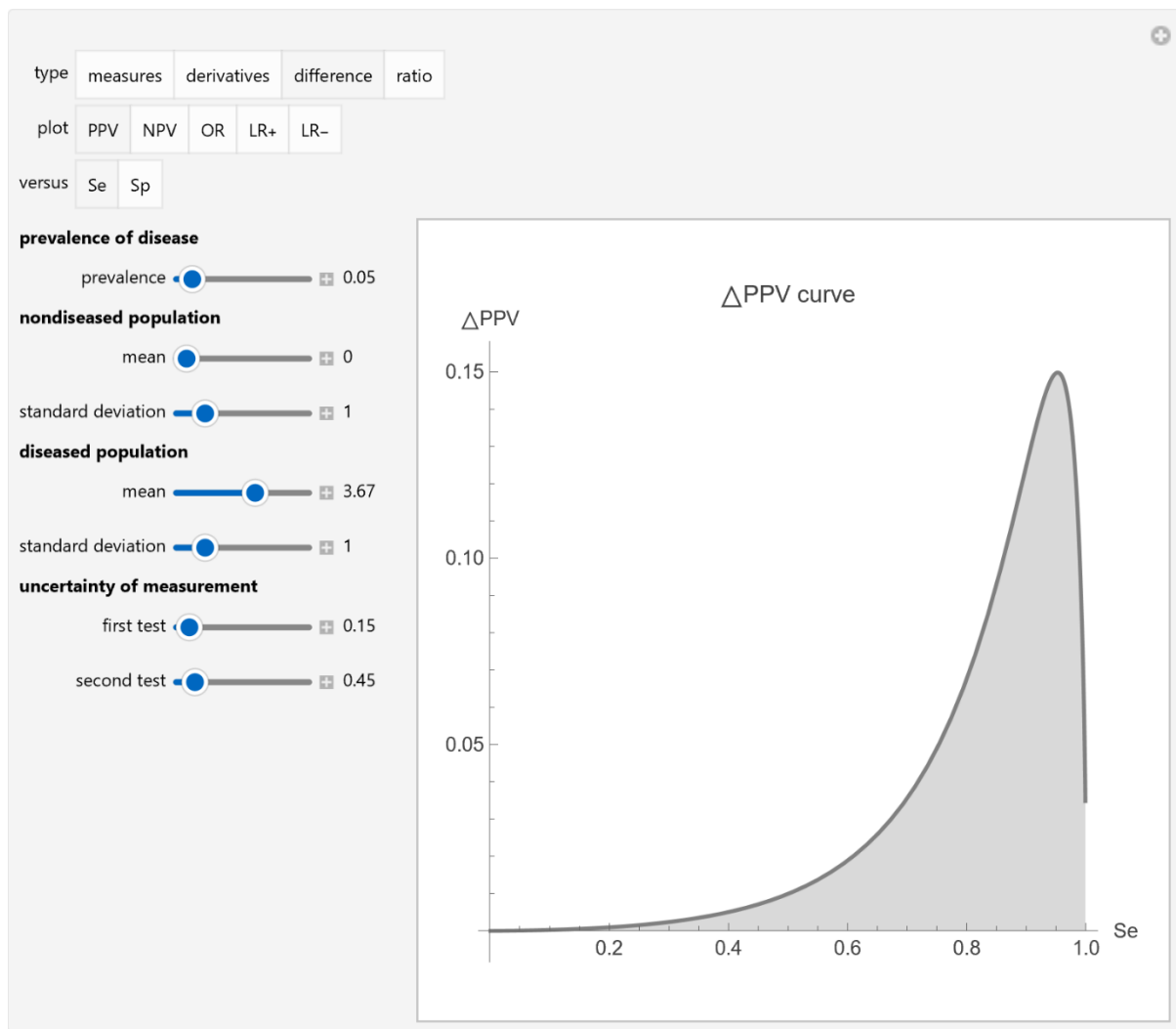


Figure 2: Positive predictive value difference between two diagnostic tests vs sensitivity curve plot, with the settings shown at the left.

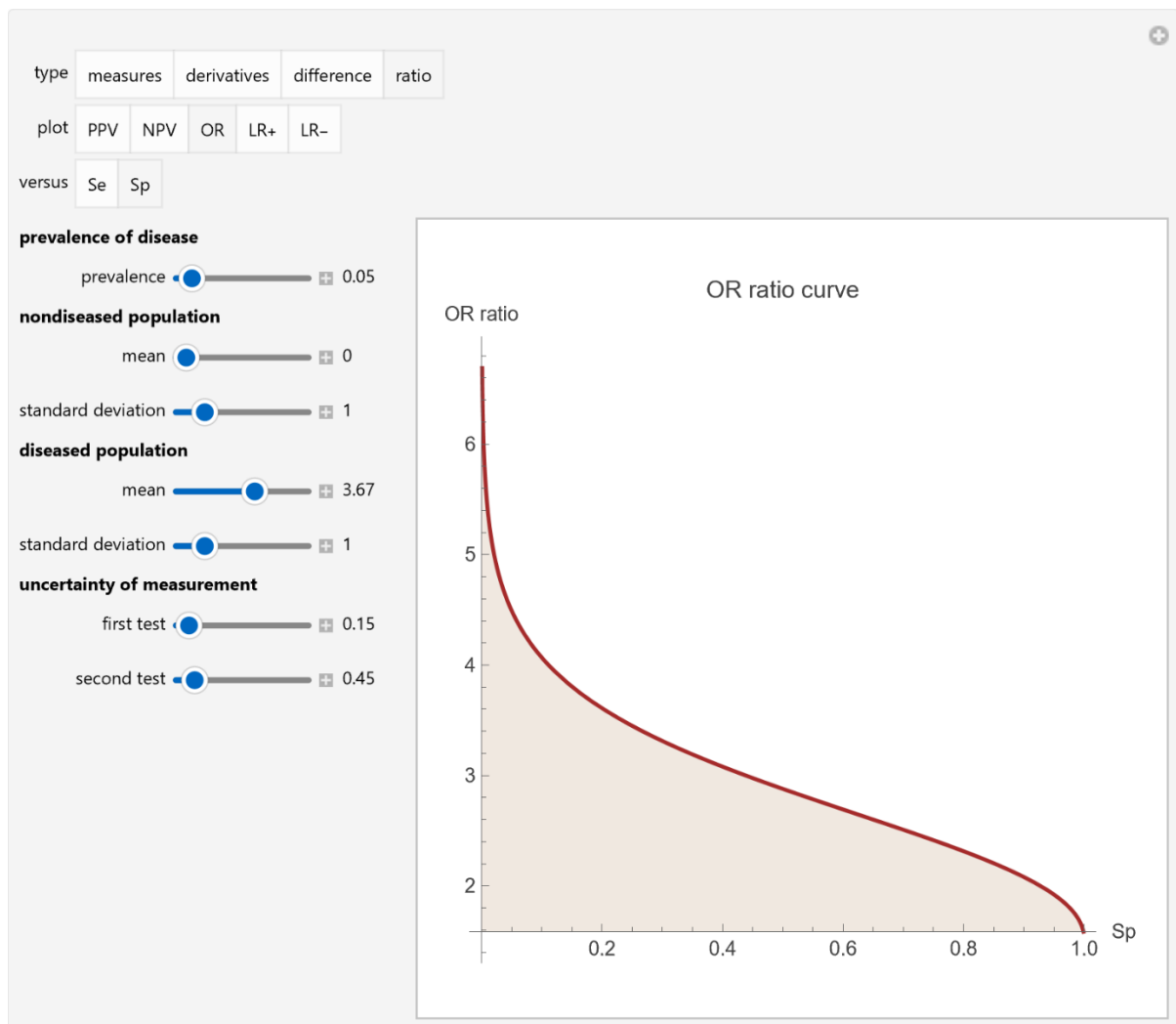


Figure 3: Ratio of the diagnostic odds ratios of two diagnostic tests plotted versus specificity, with the settings shown on the left.

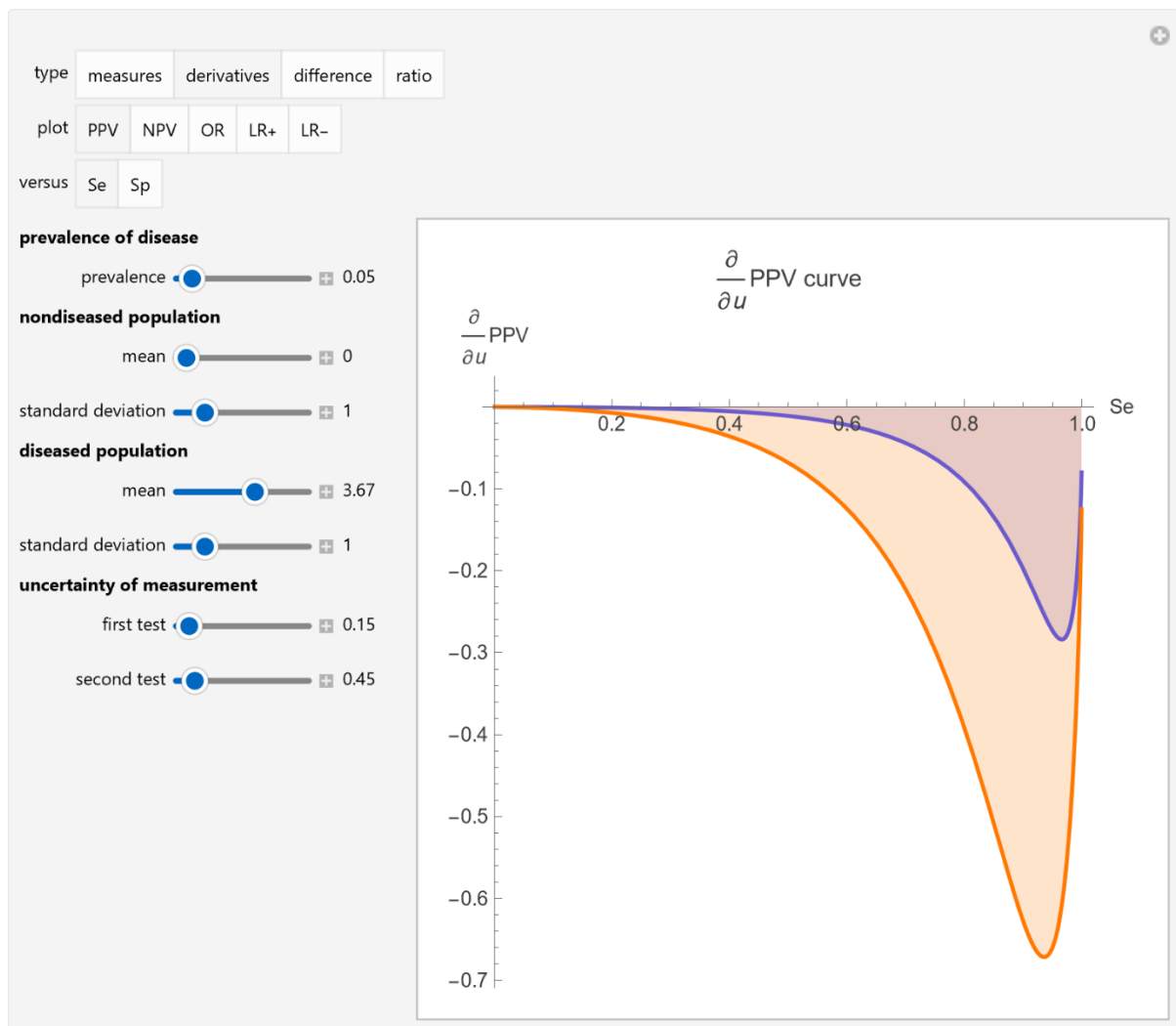


Figure 4: Partial derivative of the positive predictive value with respect to measurement uncertainty for two diagnostic tests, plotted versus sensitivity, with the first test shown in blue and the second test in orange. The settings are shown on the left.

Details

In addition to the receiver operating characteristic (ROC) curves, various measures are used in the evaluation of the clinical accuracy of a diagnostic test applied to a diseased or a nondiseased population. They can be calculated versus the sensitivity or the specificity of the test. Sensitivity is the fraction of the diseased population with a positive test, while specificity is the fraction of the nondiseased population with a negative test. In addition, if we denote by Se and Sp the sensitivity and the specificity of the combined tests, and by v the prevalence of the disease, we have:

$$PPV = \frac{Se v}{Se v + (1 - Sp)(1 - v)}$$

$$NPV = \frac{Sp (1 - v)}{Sp (1 - v) + (1 - Se)v}$$

$$OR = \frac{\frac{Se}{1 - Se}}{\frac{1 - Sp}{Sp}}$$

$$LR^+ = \frac{Se}{1 - Sp}$$

$$LR^- = \frac{1 - Se}{Sp}$$

This Demonstration could be useful in evaluating the maximum medically permissible uncertainty of measurement of a diagnostic test. For example, in the thumbnail and the snapshots the population data describe a bimodal distribution of serum glucose measurements in nondiabetic and diabetic populations [1]. The first test has a state-of-the-art performance, while the second test has a greater uncertainty.

Reference

[1] T. O. Lim, R. Bakri, Z. Morad, and M. A. Hamid. Bimodality in Blood Glucose Distribution: Is It Universal? *Diabetes Care*, **25**(12), 2002 pp. 2212–2217.

Demonstration

DOI

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Source

Programming language: Wolfram Language

Availability: The updated source notebook is available at:

<https://www.hcsl.com/Tools/Demonstrations/UncertaintyOfMeasurementAndDiagnosticAccuracyMeasures.nb>

First published in 2009 as part of the [Wolfram Demonstrations Project](https://demonstrations.wolfram.com/):

<https://demonstrations.wolfram.com/UncertaintyOfMeasurementAndDiagnosticAccuracyMeasures/>

Software Requirements

Operating systems: Microsoft Windows, Linux, Apple macOS and iOS

Other software requirements: Wolfram Player®, freely available at: <https://www.wolfram.com/player/> or Wolfram Mathematica®.

System Requirements

Processor: x86-64 compatible CPU.

System memory (RAM): 4 GB+ recommended.

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